Producer Theory

1 Technology

1. Production is a process of transforming inputs into outputs. The fundamental problem firms must contend with in this process is the **technological feasibility**. The state of technology determines and restricts what is possible in combining inputs to produce output.

2. We first concentrate on the case in which the firm produces only one output. The case of multiple outputs will be dealt with later. When there is only one output, we will use $q \in R_+$ to denote the firm's output and $x \in R_+^n$ to denote the firm's inputs.

3. We define the production function as:

   $$q = f(x).$$

   The production function captures the technology of production. It tells us how much input x is needed to produce a fixed amount of output q.

4. Assumption 3.1. The production $f : R_+^n \to R_+$, is continuous, strictly increasing, and strictly quasiconcave on R_+^n, and $f(0) = 0$.

5. The marginal product of x_i,

   $$MP_i = \frac{\partial f}{\partial x_i},$$

   tell us how many extra units of output an extra unit of x_i produces. Unlike marginal utility in consumer theory, marginal product is objective and measurable.

6. The strict quasiconcavity assumption means that any convex combination of two input vectors can produce at least as much output as one of the original two. This would be the case if we have diminishing marginal product.

7. For any fixed level of output q, the set of input vectors producing q is called q-level isoquant. This plays the same role as indifference curve in consumer theory. An isoquant traces out all the combinations of inputs that allow that firm to produce the same quantity of output.

8. The substitutability between any pair of inputs x_i and x_j is the marginal rate of technical substitution (MRTS). MRTS measures the amount of one input i the firm would require in exchange for using a little less of another input j in order to just be able to produce the same output as before. A q-level isoquant is defined as:
$$f(x) = q.$$
Let $x_j\,(x_i)$ be the amount of x_j required to keep output constant. If we differentiate the isoquant with respect to x_i, holding q constant, we get
$$\frac{\partial f}{\partial x_j}\frac{dx_j}{dx_i} + \frac{\partial f}{\partial x_i} = 0.$$
Rearranging the terms gives the slope
$$MRTS_{ij}(x) = -\frac{dx_j}{dx_i} = \frac{\frac{\partial f}{\partial x_i}}{\frac{\partial f}{\partial x_j}}.$$
Since f is strictly quasiconcave, MRTS is diminishing.

9. Elasticity of substitution: for a production function $q = f(K, L)$, the elasticity of substitution σ, measures the proportionate change in K/L relative to the proportionate change in the MRTS along the isoquant. That is ,
$$\sigma = \frac{\%\Delta\frac{K}{L}}{\%\Delta MRTS_{L,K}} = \frac{dK/L}{dMRTS}\frac{MRTS}{K/L} = \frac{d\ln(K/L)}{d\ln MRTS}.$$
The shape of the isoquants indicates the degree of substitutability.

10. Definition: we classify the returns to scale of a production function, $f(K, L)$, as follows:

Effect of Output ($m > 1$)	Returns to Scale
$f(mx) = mf(x)$	Constant
$f(mx) < mf(x)$	Decreasing
$f(mx) > mf(x)$	Increasing

11. Constant returns to scale means if you multiply all inputs by factor $m > 1$, the output is increased by factor m. Increasing returns to scale means if you multiply all inputs by a factor m, the output increases by a factor more than m. That is, you can get proportionally more as you expand. Decreasing returns to scale is just the opposite to increasing returns. A production function can have increasing returns over some range and decreasing returns over another. In fact, a lot of times there is some optimal scale in between: it is inefficient to be too small and inefficient to be too big.

12. Locally, the elasticity of scale at point x is defined by

$$\mu(x) \equiv \lim_{m \to 1} \frac{d \ln [f(mx)]}{d \ln (m)} = \frac{\sum_{i=1}^{n} f_i(x) x_i}{f(x)}.$$

1.1 Special production functions

1. Linear production function: perfect substitutes

2. Cobb-Douglas production functions

$$q = AL^{\alpha} K^{\beta},$$

   where $A, \alpha, \beta > 0$.

3. Leontief production functions

$$q = \min \big(\alpha K, \beta L \big).$$

4. CES production function

$$q = A \left[\delta L^{\frac{\sigma-1}{\sigma}} + (1-\delta) K^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

   where $A > 0, \delta \in (0,1), \sigma > 0$. σ is the elasticity of substitution.

   (a) $\sigma = +\infty$, linear production function.
   (b) $\sigma = 1$, Cobb-Douglas production function.
   (c) $\sigma = 0$, Leontief production function.

5. Homothetical production function: homothetical production produces a linear expansion path starting from the origin, optimal input ratio at various output level is constant given fixed input prices.

2 Cost Minimization

2.1 Long-run Cost minimization

1. Some cost definition

   (a) Opportunity cost: the value of a resource in its best alternative use.

   (b) Sunk costs are those unrecoverable costs that have already been incurred and the resources have no alternative use.

2. Short-run vs. long-run cost minimization

   (a) Long-run: free to vary quantities of **all** its inputs as much as it desires.

   (b) Short-run: unable to adjust the quantities of some of its inputs.

3. Long-run cost-minimization. Let w denote input prices. The cost minimization problem is defined as:

$$\min_x \; wx$$
$$s.t.\, f(x) \geq q.$$
$$\mathcal{L} = wx + \lambda (q - f(x)).$$

   The first-order conditions:

$$\frac{\partial \mathcal{L}}{\partial x_i} = w_i - \lambda f_{x_i} = 0, \; i = 1, ..., n$$
$$\frac{\partial \mathcal{L}}{\partial \lambda} = q - f(x) = 0,$$

   From the first n equations, we get for all i

$$\frac{w_i}{f_{x_i}} = \lambda.$$

4. Recall f_{x_i} is the extra output the firm can make from one extra unit input i, so $\frac{1}{f_{x_i}}$ is the amount of input i required to produce one unit of output, and $\frac{w}{f_{x_i}}$ is the cost for producing one extra unit of output using input i. A cost-minimizing firm chooses an input combination such that the cost for producing one extra unit of output is the same no matter what input mixes the firm choose to increase output.

5. Example 1: Cobb-Douglas production function

$$q = 50L^{1/2}K^{1/2}.$$

$$MRTS_{L,K} = \frac{K}{L} = \frac{w}{r} \Longrightarrow K(r,w) = \frac{w}{r}L.$$

Plugging $K(r,w)$ into the production function

$$q = 50L^{1/2}\left(\frac{w}{r}L\right)^{1/2} \Longrightarrow L = \frac{q}{50}\left(\frac{r}{w}\right)^{1/2}.$$

Similarly,

$$K = \frac{q}{50}\left(\frac{w}{r}\right)^{1/2}.$$

6. Comparative statics of change in output

   (a) The cost minimizing input combinations, as q_0 varies, trace out the **expansion path**.

   (b) If the cost minimizing quantities of labor and capital rise as output rises, labor and capital are **normal inputs**.

   (c) If the cost minimizing quantity of an input decreases as the firm produces more output, the input is called an **inferior input**.

7. Properties of Cost Function

   If f is continuous and strictly increasing, then the cost function

$$c(w,q) \equiv \min_x \; wx \quad s.t. \, f(x) \geq q$$

   is

   (a) Zero when $q = 0$.

   (b) Continuous on its domain.

   (c) For all $w >> 0$, strictly increasing and unbounded above in q.

   (d) Increasing in w.

   (e) Homogeneous of degree one in w.

   (f) Concave in w.

   (g) Shephard's lemma: $c(w,q)$ is differentiable in w at (w^0, q^0) whenever $w^0 >> 0$,

   and

$$\frac{\partial c(w^0, q^0)}{\partial w_i} = x_i\left(w^0, q^0\right), \; i = 1, ..., n.$$

2.2 Conditional Input Demands

Suppose the production function satisfies Assumption 3.1 and that the associated cost function is twice continuously differentiable. Then

1. $x(w,q)$ is homogeneous of degree zero in w.

2. The substitution matrix,

$$\begin{pmatrix} \frac{\partial x_1(w,q)}{\partial w_1} & \frac{\partial x_1(w,q)}{\partial w_n} \\ \cdot & \cdot \\ \cdot & \cdot \\ \frac{\partial x_n(w,q)}{\partial w_1} & \frac{\partial x_n(w,q)}{\partial w_n} \end{pmatrix}$$

   is symmetric and negative semidefinite. In particular, the negative semidefiniteness property implies that for all i

$$\frac{\partial x_i(w,q)}{\partial w_i} \leq 0.$$

2.3 Short-run cost-minimization

1. The firm's short-run cost minimization problem is to choose quantities of the variable inputs so as to minimize total costs, given that the firm wants to produce an output level q and under the constraint that the quantities of the fixed factors do not change.

$$\min_x \; wx + \bar{w}\bar{x}$$
$$s.t. \; f(x,\bar{x}) \geq q.$$

2. Three inputs short-run cost-minimization

$$\min_{L,M} wL + mM + r\bar{K}$$
$$s.t. \quad q = f(L,\overline{K},M)$$

   Note: L, M are the variable inputs and $wL + mM$ is the total variable cost. $\bar{K}$ is the fixed input and $r\bar{K}$ is the total fixed cost.

   Tangency condition:

$$\frac{MP_L}{w} = \frac{MP_M}{m}.$$

   Constraint:

$$q = f(L,\overline{K},M).$$

3. Suppose that $\bar{K}$ is the long run cost minimizing level of capital for output level q^*. Then when the firm produces q^*, the short-run demands for L and M must yield the long-run cost minimizing levels of L and M

4. Example

$$q = K^{1/2}L^{1/4}M^{1/4}, \qquad r = 2, w = 16, m = 1.$$

The short-run cost-minimization condition when $q = 16$, $\bar{K} = 32$:

(a) Short-run Tangency condition:

$$\frac{MP_L}{MP_M} = \frac{M}{L} = \frac{w}{m} = \frac{16}{1} \Longrightarrow M = 16L.$$

Constraint

$$16 = 32^{1/2}L^{1/4}\left(16L\right)^{1/4},$$

which gives $L = 2$, and $M = 16L = 32$. Thus when $q = 16, \bar{K} = 32$, the short-run cost minimization inputs combination is

$$L = 2, M = 32.$$

(b) Long-run cost minimization when $q = 16$:
Two tangency condition

$$\frac{MP_L}{MP_M} = \frac{M}{L} = \frac{w}{m} = \frac{16}{1} \Longrightarrow M = 16L,$$

$$\frac{MP_L}{MP_K} = \frac{K}{2L} = \frac{w}{r} = \frac{16}{2} \Longrightarrow K = 16L$$

Constraint

$$16 = \left(16L\right)^{1/2}L^{1/4}(16L)^{1/4} \Longrightarrow 16^{\frac{3}{4}}L = 16 \Longrightarrow L = 16^{\frac{1}{4}} = 2.$$

Plugging $L = 2$ into tangency condition

$$M = 16L = 32, \qquad K = 16L = 32.$$

Hence when $\bar{K} = 32$ is the long-run cost minimizing level of capital for the output level $q = 16$, the short-run cost minimization yields the long-run cost minimizing levels of L and M.

5. For the previous example, the long-run cost given input prices $r = 1, w = 16, m = 1$ is
$$c(q) = 8q.$$
The short-run cost fixing $\bar{K} = 32$ is
$$sc(q, \bar{K}) = 64 + \frac{q^2}{4}.$$
Clearly $sc(q, \bar{K}) \geq c(q)$ for all $q \geq 0$.

3 Profit Maximization

1. To find the optimal output
$$\max_{\{q\}} pq - c\,(w, q)$$
The first order condition implies that
$$p = \frac{\partial c}{\partial q} \tag{1}$$
and the second order condition is that
$$\frac{\partial^2 c}{\partial q^2} > 0.$$
By the first order conditions, we can get the firm's output supply function, $q^* = q(p, w)$.

2. To find the optimal inputs
$$\max_{\{x\}} pf\,(x) - wx$$
Using the first order condition we have
$$p\frac{\partial f}{\partial x_i} = w_i, \text{for } i = 1, ..., n.$$
The first-order conditions mean that the marginal revenue product (MRP) of each input i should equal the input price. MRP gives the rate at which revenue increases per additional unit of input i employed. The first-order conditions will give us the input demand function of the firm, $x^* = x(p, w)$.

3. Theorem 3.7. If f satisfies Assumption 3.1, then for $p \geq 0$ and $w \geq 0$, the profit function
$$\pi\,(p, w) \equiv \max pq - wx$$
when well-defined, is continuous and

(a) Increasing in p.

(b) Decreasing in w.

(c) Homogeneous of degree one in (p, w).

(d) Convex in (p, w).

(e) Differentiable in $(p, w) >> 0$. Moreover, (Hotelling's lemma),

$$\frac{\partial \pi}{\partial p} = q,$$
$$-\frac{\partial \pi}{\partial w_i} = x_i.$$

4. Theorem 3.8. Suppose π is well-defined and twice continuously differentiable, then

   (a) q and x_i are homogeneity of degree zero.

   (b) The substitution matrix

$$\begin{pmatrix} \frac{\partial q}{\partial p} & \frac{\partial q}{\partial w_1} & \cdots & \frac{\partial q}{\partial w_n} \\ -\frac{\partial x_1(p,w)}{\partial p} & -\frac{\partial x_1(p,w)}{\partial w_1} & \cdots & -\frac{\partial x_1(p,w)}{\partial w_n} \\ \cdots & \cdots & \cdots & \cdots \\ -\frac{\partial x_n(p,w)}{\partial p} & -\frac{\partial x_n(p,w)}{\partial w_1} & \cdots & -\frac{\partial x_n(p,w)}{\partial w_n} \end{pmatrix}$$

   is symmetric and positive semidefinite. In particular, $\frac{\partial q}{\partial p} \geq 0$ and $\frac{\partial x_i(p,w)}{\partial w_i} \leq 0$ for all i.

5. Example: Price P, input prices (w_1, w_2)

$$q = x_1^{\alpha} x_2^{\beta}$$

   where $\alpha, \beta > 0$. The first order condition gives

$$P\alpha x_1^{\alpha-1} x_2^{\beta} - w_1 = 0, \tag{2}$$

$$P\beta x_1^{\alpha} x_2^{\beta-1} - w_2 = 0, \tag{3}$$

$$q - x_1^{\alpha} x_2^{\beta} = 0. \tag{4}$$

   The first two conditions imply that $\alpha x_2/\beta x_2 = w_1/w_2$ and thus, $x_2 = (\beta w_1/\alpha w_2)x_1$. Plugging it back into (4) yields

$$x_1 = q^{\frac{1}{\alpha+\beta}} \left(\frac{\alpha w_2}{\beta w_1}\right)^{\frac{\beta}{\alpha+\beta}}, \quad x_2 = q^{\frac{1}{\alpha+\beta}} \left(\frac{\beta w_1}{\alpha w_2}\right)^{\frac{\alpha}{\alpha+\beta}}.$$

Plugging the two conditional input demands into (2) gives

$$P\alpha q^{\frac{\alpha+\beta-1}{\alpha+\beta}}\left(\frac{\alpha w_2}{\beta w_1}\right)^{\frac{-\beta}{\alpha+\beta}} = w_1 \Longrightarrow q^{\frac{\alpha+\beta-1}{\alpha+\beta}} = P^{-1}w_1^{\frac{\alpha}{\alpha+\beta}}w_2^{\frac{\beta}{\alpha+\beta}}\alpha^{\frac{-\alpha}{\alpha+\beta}}\beta^{\frac{-\beta}{\alpha+\beta}} \Longrightarrow$$

$$\begin{aligned} q &= P^{\frac{\alpha+\beta}{1-\alpha-\beta}}w_1^{\frac{\alpha}{\alpha+\beta-1}}w_2^{\frac{\beta}{\alpha+\beta-1}}\alpha^{\frac{\alpha}{1-\alpha-\beta}}\beta^{\frac{\beta}{1-\alpha-\beta}}, \\ x_1(P,w) &= P^{\frac{1}{1-\alpha-\beta}}w_1^{\frac{1-\beta}{(\alpha+\beta-1)}}w_2^{\frac{\beta}{(\alpha+\beta-1)}}\alpha^{\frac{1-\beta}{(1-\alpha-\beta)}}\beta^{\frac{\beta}{1-\alpha-\beta}}, \\ x_2(P,w) &= P^{\frac{1}{1-\alpha-\beta}}w_1^{\frac{\alpha}{(\alpha+\beta-1)}}w_2^{\frac{1-\alpha}{(\alpha+\beta-1)}}\alpha^{\frac{\alpha}{(1-\alpha-\beta)}}\beta^{\frac{1-\alpha}{1-\alpha-\beta}}. \end{aligned}$$

Plugging $q(P,w)$ back into the conditional input demand gives $x_i(P,w)$. And the profit function

$$\pi(P,w) = Pq(P,w) - wx(P,w).$$

Note that when $\alpha+\beta<1$,

$$\frac{dq(P,w)}{dP} > 0 \qquad \frac{\partial x_i(P,w)}{\partial w_i} < 0.$$

6. When $\alpha+\beta=1$ or $\alpha+\beta>1$, $q(P,w)$ and $x_i(P,w)$ are undefined, therefore, the profit function is not defined. To show this, the profit function defined as before:

$$\pi(p,w) \equiv \max pf(x) - wx.$$

Suppose that x' and $f(x')$ maximize profits at p and w. With increasing returns,

$$f(tx') > tf(x') \text{ for all } t>1.$$

Thus,

$$pf(tx') - wtx' > ptf(x') - wtx' \geq pf(x') - wx' \text{ for all } t>1.$$

This says higher profit can always be had by increasing inputs in proportion $t>1$, which contradicts our assumption that x' and $f(x')$ maximized profit.

7. Note that in the special case of constant returns, no such problem arises if the maximal level of profit happens to be 0. In that case, though, the scale of the firm's operation is indeterminate because $(f(x'),x')$ and $(tf(x'),tx')$ give the same level of 0 profits for all $t>0$.

4 Multiple outputs

1. Another way to represent the technology constraint is to think of the firm as having a **production possibility set**, $Y \subset R^n$, where each vector $y = (y_1, y_2, ...y_n) \in Y$ is a **production plan**, where $y_i > 0$ means outputs and $y_j < 0$ means inputs.

2. A production plan $y \in Y$ is **technological efficient** if there is no $\widehat{y} \in Y$ such that $\widehat{y} > y$. A production plan is **economically efficient** if it maximizes profits $\pi = py$ of a given price vector p.

3. Economic efficiency implies technological efficiency. No matter what the prices of inputs and outputs are, the firm needs to achieve technological efficiency in order to achieve profit maximization. We can think of the firm's choice problem in two steps: first, the firm has to achieve technological efficiency; second, the firm obtain economic efficiency on the efficient production set or **the production possibility frontier (PPF).** The PPF is defined to be the set of all technologically efficient production plans.

4. Formally, given a production plan $y = (y_1, y_2, ...y_n) \in Y$, there is a twice differentiable function $G : R^n \to R$ that defines the production possibility set: $Y \equiv \{y \in R^n \,|G(y) \leq 0\}$.

5. Assumption 1. Strict Monotonicity. That is, $G_{y_i} > 0, \forall i, y$. By this assumption, since $G_{y_i} > 0$, any y such that $G(y) < 0$ is technologically inefficient: one can increase the production (or decrease the use) of one product y_i while keeping others constant (some y_i are inputs). Therefore, $\{y \in R^n \,|G(y) = 0\}$ is the technologically efficient production set and is thus called the production possibility frontier (PPF).

6. We define the marginal rate of transformation (MRT) as

$$MRT_{ij} \equiv -\frac{\partial y_i}{\partial y_j} = \frac{G_{y_j}}{G_{y_i}} > 0.$$

   MRT is the slope the PPF.

7. Assumption 2. G is quasi-convex. The firm's problem is then:

$$\begin{aligned} \pi(p) &\equiv \max_{y \in R^n} py \\ s.t.\ G(y) &\leq 0. \end{aligned}$$

   Since technological efficiency is necessary for economic efficiency (profit maximization), the above problem is equivalent to

$$\pi(p) \equiv \max_{y \in R^n} py$$
$$s.t.\ G(y) = 0.$$